

BIOINFORMATICS BOOTCAMP

ASSESSMENT 02

PART I

Question 1

A molecular biologist has been provided with only a gene accession number from a published research article and needs to retrieve the corresponding DNA sequence, protein sequence, associated literature, and gene information. Describe the step-by-step approach you would follow using NCBI resources. Explain which databases and tools you would access during this process and why.

Answer:

1. I will first visit the **NCBI** home page and, using the search bar, I will enter the gene accession number. I will use the **Nucleotide (GenBank) database** to search for the accession number on the NCBI. The results of the search will provide me with records that correspond to the accession number. The reason for using the Nucleotide database is that it is the primary repository for DNA and RNA sequences.
2. I will then have access to and be able to view the complete nucleotide sequence in **GenBank** or **FASTA** format. Through the GenBank format, I will have access to the information about the organism, the gene location, the DNA sequence, the coding sequence, and the functional annotations. I will then download the **GenBank** format of this record to my computer.
3. To obtain the protein sequence, I will stay within the nucleotide record and locate the **Coding Sequence (CDS)** feature. This will provide me with the protein accession number. I will follow the linked accession number to the **NCBI Protein database**. This record will provide me with the name of the protein, the site, functions, conserved domains, and the amino acid sequence. I will then download the amino acid sequence in the **FASTA** format to my computer.
4. I will copy the sequence and use **BLASTn** and **BLASTp** to verify the nucleotide sequence and protein sequence, respectively. I will then compare against the NCBI databases to identify any sequence identity, homologous genes, and conserved regions.
5. To find out about the gene information, I will use **NCBI Gene** because it collects comprehensive information about a gene from multiple databases. The Gene record will provide me with the official symbol of the gene and its full name, the structure and

functions, biological pathways, expression information, orthologs and homologs, as well as links to related NCBI resources.

6. From the Gene record, I will follow the links to **PubMed** because it provides the biomedical literature. From PubMed, I will review the original publication describing the gene, functional studies, the diseases associated with this particular gene, and access full recent research articles using **PubMed Central (PMC)** because PMC provides free full-text articles.

Question 2

A structural biologist is interested in studying the three-dimensional structure of the human insulin protein to understand how mutations may affect its function. Which database would you recommend for this purpose? Explain how you would retrieve the structure and describe the information available within the database that would support the research.

Answer:

The **Protein Data Bank (PDB)** is the most appropriate database for 3-D protein structure since it provides the atomic coordinates of the protein.

1. I will first visit the **RCSB PDB** website that allows for structural and functional studies, where I will have free access to experimentally determined macromolecular proteins like the human insulin protein.
2. I will then enter “human insulin” into the search bar, which will provide me with more than one insulin structure.
3. I will select the most appropriate protein structure that is linked to my objective of understanding how mutations may affect its function. To come up with the most appropriate structure, I will take into consideration the experimental method used, either X-ray crystallography, NMR spectroscopy, or Cryo-Electron Microscopy; the resolution (because to have access to more structural detail, a higher resolution is needed); whether the structure contains wild-type or mutant insulin structures; and the presence of bound ligands or receptors.
4. Using the integrated 3-D molecular viewer available in the PDB, I will visualize the secondary structural elements like the α -helices and β -sheets, disulfide bonds, which are specifically important for insulin stability and biological activity, active or binding regions, and individual amino acid residues.

5. I will download then download the 3-D structure and use PyMOL software to further analyze the structure.

Question 3

Imagine you have been assigned a protein with no prior knowledge except its accession ID. Design a complete workflow describing how you would retrieve, analyze, and interpret all available information about this protein using NCBI, UniProt, and the Protein Data Bank (PDB). Explain the purpose of each step and how the information from one database complements the others.

Answer:

Step 1 - Identify the protein using the accession number: NCBI Protein

I will open NCBI, select the Protein database, and enter the protein accession ID into the NCBI search bar. I will then open the corresponding protein record, where I will have access to the name of the protein, the organism, the amino acid sequence, the length of the protein, the coding gene, the functions, and cross-reference to other NCBI databases. This is to obtain the primary protein sequence and verify that the accession number corresponds to the correct protein.

Step 2 - Retrieval of Gene Information: NCBI Gene

From the protein record, I will follow the link to the Gene Database to retrieve the gene information. This will give me access to the official gene symbol and name, gene structure, chromosomal location, expression information, biological function, orthologs and homologs, and the diseases associated with it. This information will give me a clear understanding of the genetic context of the protein and the relationship between the protein and its encoding gene.

Step 3 - Assess associated literature: PubMed

I will use the links provided in the gene records to access relevant publications to know the protein function, biological pathways, mutational analyses, disease associations, and recent discoveries. Through this, I will have the current knowledge about the protein and have access to experimental validation of that protein.

Step 4 - Perform Sequence Similarity Analysis: BLASTp

I will perform sequence similarity analysis using NCBI BLASTp to confirm protein identity, identify homologous proteins because they can suggest functions for proteins that have not been

experimentally characterized, determine evolutionary relationships, and predict function based on sequence similarity.

Step 5 - Retrieve comprehensive protein annotation: UniProt

I will search the accession number using UniProt to obtain a comprehensive protein annotation to have access to protein function, catalytic activity, biological processes, molecular function, cellular location, protein domains, post-translational modifications, sequence variants, protein interactions, isoforms, and cross-references to numerous biological databases.

Step 6: Study Protein Domains and Functional Sites: UniProt

UniProt has links to domain resources, so I will review them to identify functional domains, active sites, binding residues, signal peptides, transmembrane regions, and conserved motifs.

Step 7: Retrieve 3D Structure: Protein Data Bank (PDB)

I will then retrieve the 3D structure using PDB by searching the accession number to have access to the experimentally determined structure. I will visualize the overall protein folding, α -helices and β -sheets, tertiary structure, ligand-binding sites, protein complexes, and mutation sites. I will then download the structure in PDB or mmCIF format for further analysis.

Step 8: Interpret Structure–Function Relationships: UniProt and PDB

I will compare the structural features of the protein in PDB to the functional annotations in UniProt to understand how the protein performs its biological role. I will check whether active sites correspond to conserved residues, how domains contribute to function, where disease-causing mutations are located, the structural basis of ligand binding, and stability-related structural features.

Step 9: Integrate the Information: NCBI, UniProt, PDB

I will combine my findings from all databases to generate a complete protein profile about protein identity, amino acid sequence, encoding gene, biological function, molecular function, cellular location, conserved domains, structural characteristics, evolutionary relationships, protein interactions, associated diseases, and relevant scientific literature.

PART II

Question 4

Choose one biological sequence (DNA or Protein) from NCBI, UniProt, or the Protein Data Bank (PDB). Include the following in your submission:

- Screenshot of the retrieved database record: UniProt

The screenshot shows the UniProtKB search results for the query 'protein_name:angiotensin'. The interface includes a top navigation bar with options like 'Name(s) and synonym(s) of the protein', 'Sequence search', 'ID mapping', 'SPARQL', and 'UniProtKB'. The search bar contains the query 'protein_name:angiotensin' and a 'Search' button. Below the search bar, the results are displayed as a table with 9,996 results. The table has columns for Entry, Entry Name, Protein Names, Gene Names, Organism, and Length. The first few entries are listed below:

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
P01019	ANGT_HUMAN	Angiotensinogen[...]	AGT, SERPINA8	Homo sapiens (Human)	476 AA
Q9BYF1	ACE2_HUMAN	Angiotensin-converting enzyme 2[...]	ACE2, UNQ868/PRO1885	Homo sapiens (Human)	805 AA
P50052	AGTR2_HUMAN	Type-2 angiotensin II receptor[...]	AGTR2	Homo sapiens (Human)	363 AA
P30556	AGTR1_HUMAN	Type-1 angiotensin II receptor[...]	AGTR1, AGTR1A, AGTR1B, AT2R1, AT2R1B	Homo sapiens (Human)	359 AA
P12821	ACE_HUMAN	Angiotensin-converting enzyme[...]	ACE, DCP, DCP1	Homo sapiens (Human)	1,306 AA
P01015	ANGT_RAT	Angiotensinogen[...]	Agt, Serpina8	Rattus norvegicus	477 AA

The screenshot displays the UniProtKB entry for P01019, ANG_T_HUMAN. The left sidebar lists various categories like Function, Names & Taxonomy, and Subcellular Location. The main content area provides key information: Protein is Angiotensinogen, Gene is AGT, Status is UniProtKB reviewed (Swiss-Prot), and Organism is Homo sapiens (Human). It also notes 476 amino acids and evidence at the protein level. The 'Function' section describes it as an essential component of the renin-angiotensin system (RAS), a potent regulator of blood pressure, body fluid and electrolyte homeostasis. Below this, a snippet of the protein description is visible: 'Angiotensin-2 Acts directly on vascular smooth muscle as a potent vasoconstrictor, affects cardiac contractility and heart rate through its...'.

- **Accession ID or Entry ID of the selected sequence:** P01019
- **Name of the Gene/Protein:** AGT/Angiotensinogen
- **Source Organism:** Homo sapiens (human)
- **A brief description (150–250 words):**

Angiotensinogen is a plasma glycoprotein encoded by the **AGT** gene in *Homo sapiens*. It circulates in the bloodstream as the precursor of the angiotensin peptides that potentially regulate blood pressure, body fluid, and electrolyte homeostasis via the Renin-Angiotensin-Aldosterone system (RAAS). It is cleaved into 8 chains: Angiotensin I, Angiotensin II, Angiotensin III, Angiotensin IV, Angiotensin 1-9, Angiotensin 1-7, Angiotensin 1-5, and Angiotensin 1-4.

In the instance of low blood pressure, the enzyme renin cleaves angiotensinogen to produce angiotensin I, which is subsequently converted into angiotensin II, a potent vasoconstrictor, by angiotensin-converting enzyme (ACE). Angiotensin II plays an integral role in regulating systemic arterial pressure by acting directly on vascular smooth muscle. It acts on the sympathetic nervous system and affects heart rate and cardiac contractility. It causes vasoconstriction, stimulates aldosterone secretion, and promotes sodium and water retention by the kidneys, thereby raising blood pressure. It is required for maintaining cardiovascular and renal homeostasis. Abnormal expression or mutations in the AGT gene have been associated with hypertension, cardiovascular disease, kidney disorders, and pregnancy complications such as pre-eclampsia. Consequently,

angiotensinogen and its downstream pathway are important therapeutic targets in the treatment of hypertension and heart failure.

I chose the UniProt database because of the comprehensive and curated information it provides. It provides the amino acid sequence, together with detailed information on protein function, biological pathways, subcellular location, disease associations, sequence features, and cross-references to other databases, making it an excellent resource for studying the structure and function of angiotensinogen.